

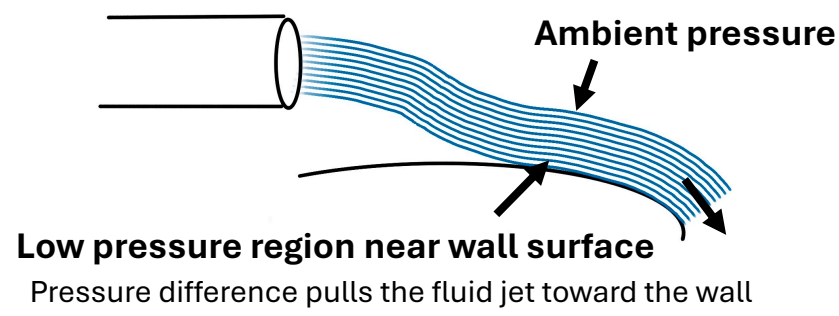
CFD STUDY OF BUOYANCY AND MOMENTUM INFLUENCES ON THE COANDA EFFECT IN A LAMINAR LIQUID SYSTEM

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INTRODUCTION

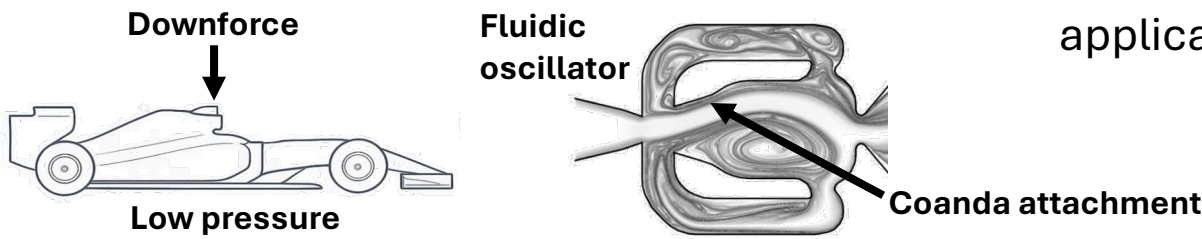
What is the Coanda effect?

The tendency of a fluid jet to attach to an adjacent curved surface rather than continuing in a straight line.



Why is it important?

- The Coanda effect has been studied widely in turbulent applications, from aircraft and HVAC to Formula 1 racing.
- However, it has not been studied much in laminar liquid systems, where buoyancy forces add complexity to the jet attachment behaviour.



What is this project doing?

- Explores the Coanda effect in a laminar liquid system.
- Investigating the relative influence of momentum and buoyancy forces on the fluid jet attachment.
- Identifies the conditions for the Coanda effect presence and directions.
- Developed CFD in COMSOL, validated the models against experimental results.
- The CFD models provide a framework for future applications of the Coanda effect.

METHODOLOGY

Experiment

- Latex dispersion was injected into a vertical tank containing a saltwater solution.
- Varying the latex injection flowrate to investigate momentum, and adding salt to increase the solution density and alter buoyancy forces.

Validation Assessment

- Compare CFD with experiment
- Iterative convergence
- Mass conservation
- Mesh size analysis
- Sensitivity analysis
- Consistency

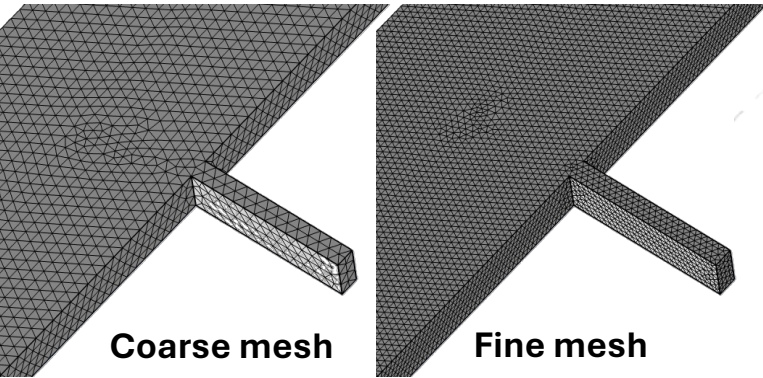


Figure 1: CFD mesh – coarse (left) and fine (right)

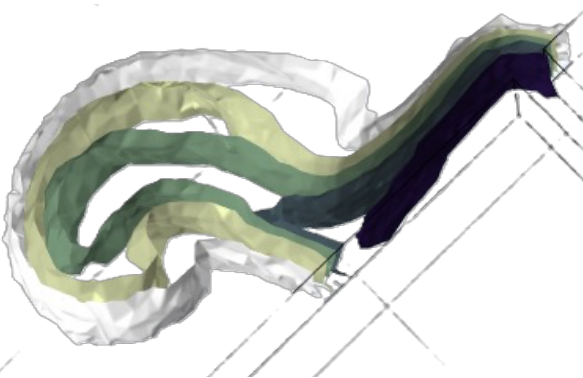


Figure 2: CFD 3D simulation – downward attachment at 1ml/min

Computational Fluid Dynamics (CFD)

- Developed 2D and 3D CFD models replicating the experimental geometry, flow conditions, and fluid phases.
- Coupled interfaces and shared equations were used to simultaneously model fluid flow, species transport, and density changes (See Fig.3).
- Latex was modelled as a passive tracer, eliminating the need for a computationally expensive three-phase model.

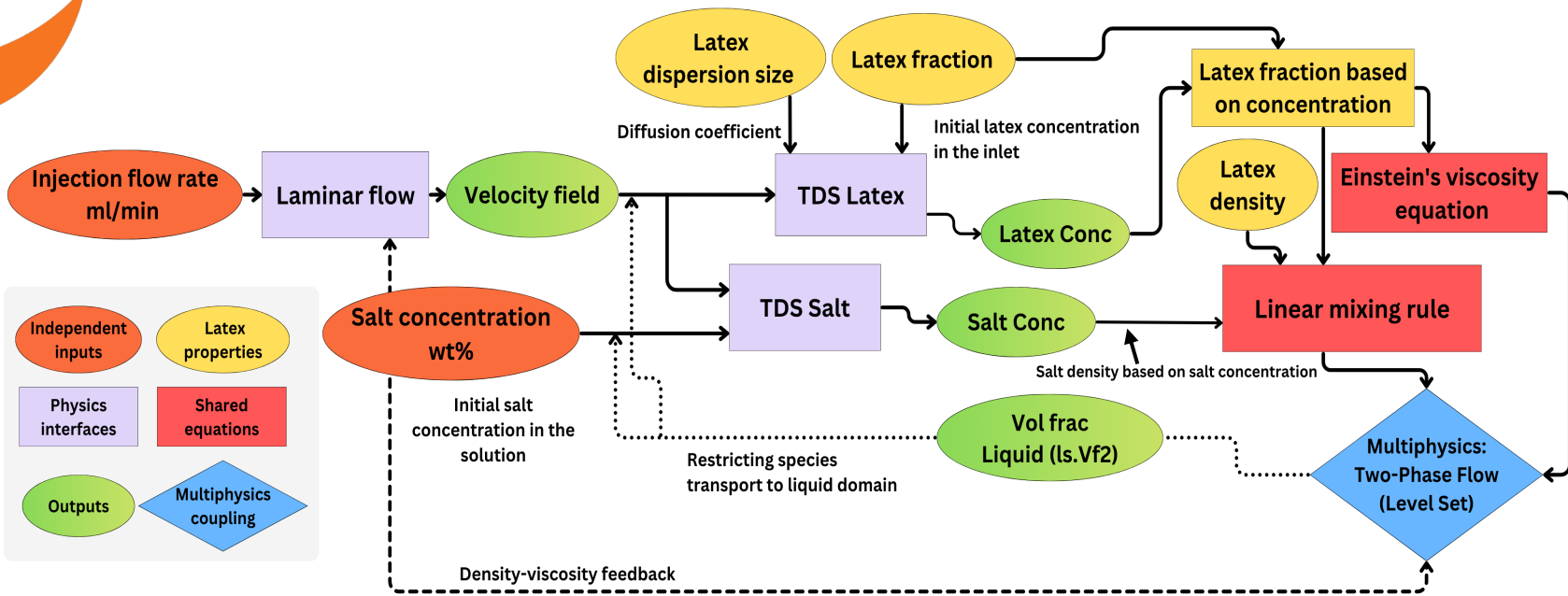
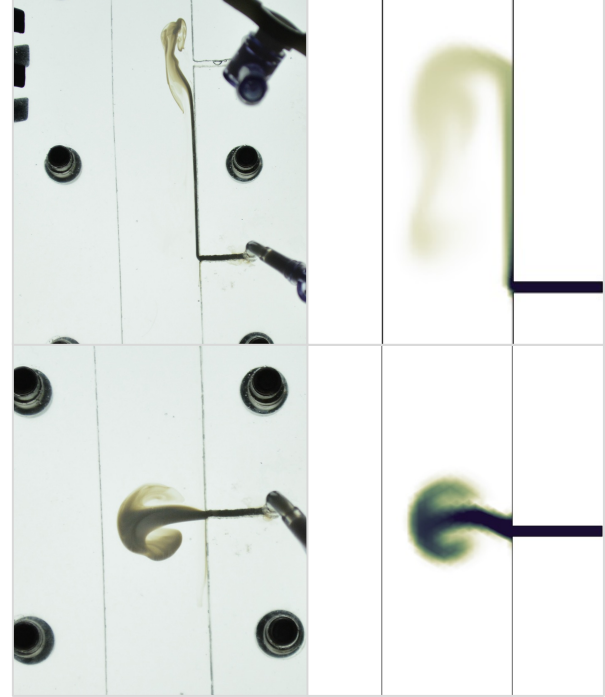


Figure 3: Flow diagram demonstrating the logic behind the CFD model, how variables and physics interfaces interact with each other.

RESULTS



81.8% (at minimum)
Agreement: CFD vs Experiment
(direction and bifurcation behaviour)

Mesh Refinement (coarse → fine)
(tested on selected conditions)

Agreement level 81.8% → 87.9%
(direction and bifurcation)

Attachment thickness improved

Crossover point: Jet transitions to upward attachment above 0.3wt% salt

Figure 4: Experiment (left) vs CFD 2D (right) — upward attachment (top) and no coanda (bottom)

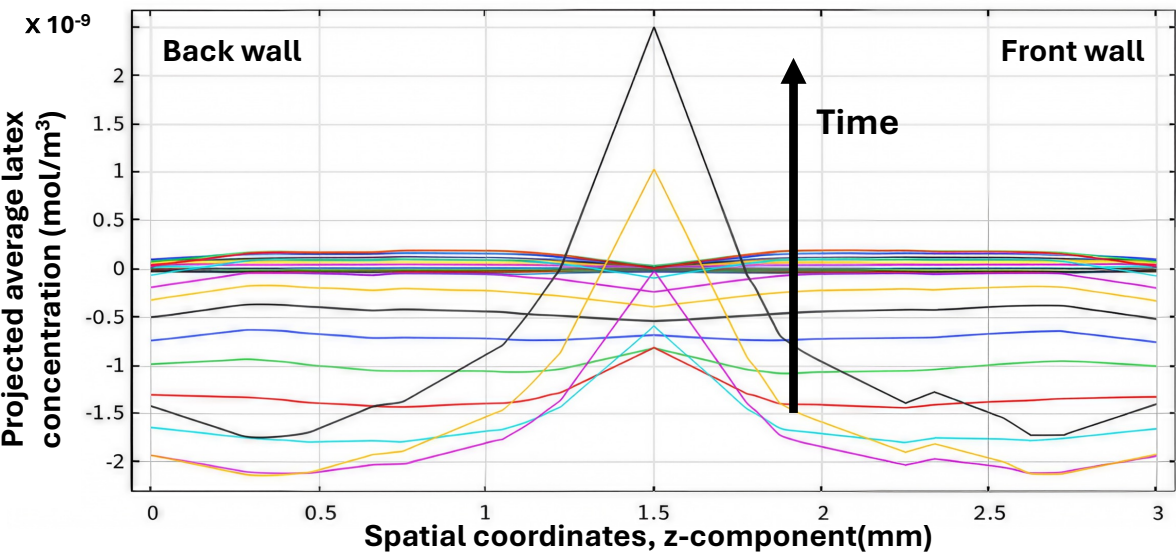


Figure 5: Projected latex concentration profile across the tank width (z-direction) obtained from the 3D model at 0.1ml/min, plotted at every time step for 25 seconds. The near-symmetrical 3D profiles show little variation between the front and the back walls, indicating that 2D captured the majority of the Coanda behaviour.

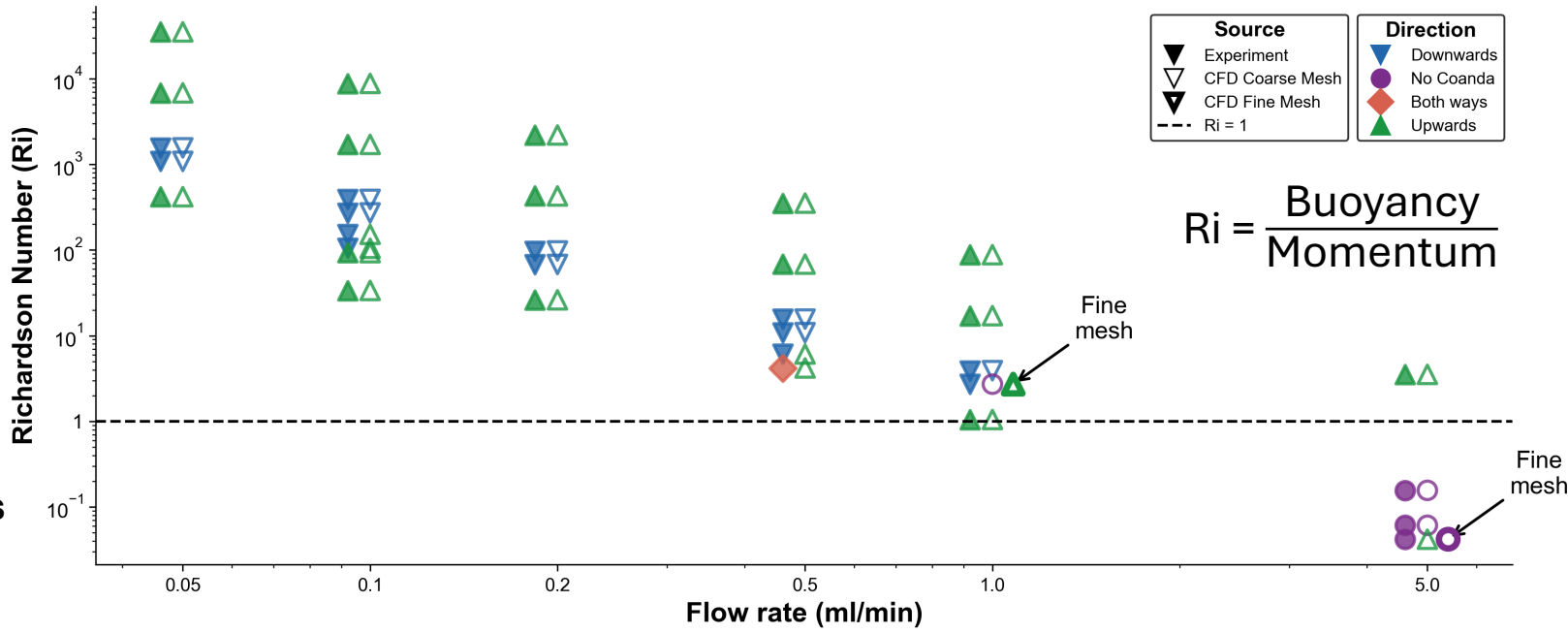


Figure 6: Richardson number plot vs flow rate for experimental, CFD coarse mesh, and CFD fine mesh results. Coanda effect observed mostly above $Ri = 1$, suggesting the critical threshold lies below $Ri = 1$ for laminar liquid systems, confirming buoyancy dominates jet attachment.

CONCLUSIONS

- The CFD model proved effective at predicting the presence and direction of the Coanda effect, with **81.8% agreement** with experimental results.
- For a laminar liquid system, buoyancy controls attachment, as the Coanda effect was mostly present at **$Ri > 1$** .
- A crossover point was consistently predicted by the CFD at **0.3 wt% salt**.
- The passive tracer method was proved to be effective for modelling latex while saving computational costs.

FUTURE WORK

- Further exploring the effect of mesh refinement on model accuracy.
- Implementing adaptive mesh strategies, using high-resolution mesh only near the wall surface.
- Identify the critical Richardson number threshold and crossover point by running experiments/simulations at a higher resolution of salt concentrations.

References

- Felix. Upgrade Motorsport [Internet]. 2025 [cited 2026 Apr 20]. The Invisible Force in Motorsport – An Introduction to F1 Aerodynamics by Guest Writer Felix. Available from: <https://www.upgrademotorsport.co.uk/post/the-invisible-force-in-motorsport-an-introduction-to-f1-aerodynamics-by-guest-writer-felix>
- Sieber M. Flow field of a fluidic oscillator [Internet]. 2015 [cited 2026 Apr 21]. Located at: Own work. Available from: https://commons.wikimedia.org/wiki/File:Flow_field_of_a_fluidic_oscillator.svg